BIG IDEAS:

- Any number (except 0) **to the power of 0** is equal to **1** $a^0 = 1 \quad (a \neq 0)$
- a^{-b} is equal to the **reciprocal** of a^b $a^{-b} = \frac{1}{a^b}$
- A number written in **scientific notation** has a **coefficient** (with one digit in front of the decimal point) **multiplied by a power of 10**

EXAMPLE 1

6.23×10^6

$= 6.230000$

$= 6\,230\,000$

EXAMPLE 2

4.872×10^{-4}

$= 0004.872$

$= 0.0004872$

EXAMPLE 3

$4\,567\,000\,000$

$= 4.567 \times 10^9$

EXAMPLE 4

5^0

$= 1$

EXAMPLE 5

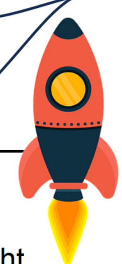
5^{-2}

$= \frac{1}{5^2}$

$= \frac{1}{25}$

MORE EXAMPLES

Things to remember…



THINKING BEYOND…

1. The diameter of the Milky Way galaxy is approximately 9.5×10^{17} kilometres. The speed of light is approximately 3.0×10^8 metres per second. How long does it take light to travel from one side of the Milky Way to the other?
2. The mass of a proton is $1.67262192 \times 10^{-24}$ grams. How many protons are needed to total a mass of 1 kilogram?
3. Use a calculator to evaluate 0^0. Suggest an explanation for the result.
4. Determine the exact value of $2^{-3} + 3^{-2}$.
5. Determine the value of the infinite sum $3^{-1} + 3^{-2} + 3^{-3} + 3^{-4} + \ldots$. How can you justify your answer?